NEET (UG) 2027 — Model Question Paper 2

Duration: 3 hours · Maximum marks: 720 · 180 questions (Physics 45, Chemistry 45, Botany 45, Zoology 45)

1. All questions are compulsory multiple-choice with one correct option.
2. +4 for each correct answer, −1 for each wrong answer, 0 if unattempted.
3. Take g = 10 m/s² unless stated otherwise.
4. Answer key is at the end of the paper.

Physics

1. The dimensional formula of impulse is

(1) $[MLT^{-2}]$ **(2)** $[ML^2T^{-1}]$
(3) $[MLT^{-1}]$ **(4)** $[ML^{-1}T]$

2. If the side of a square is measured with 2% error, the percentage error in its area is

(1) 4% **(2)** 2%
(3) 1% **(4)** 8%

3. A car starts from rest with acceleration 2 m/s². Distance covered in 5 s is

(1) 10 m **(2)** 50 m
(3) 20 m **(4)** 25 m

4. A projectile is thrown at 20 m/s at 30° to the horizontal (g = 10 m/s²). Maximum height is

(1) 10 m **(2)** 5 m
(3) 20 m **(4)** 2.5 m

5. For uniform circular motion, the acceleration is

(1) v^2/r towards the centre **(2)** v^2/r tangential
(3) zero **(4)** vr towards the centre

6. The momentum of a 2 kg body with kinetic energy 100 J is

(1) 10 kg m/s **(2)** 200 kg m/s
(3) 20 kg m/s **(4)** 40 kg m/s

7. Rocket propulsion is based on conservation of

(1) energy **(2)** mass
(3) angular momentum **(4)** linear momentum

8. For a stone whirled in a vertical circle of radius r, the minimum speed at the top for the string to remain taut is

(1) √(2gr) **(2)** √(gr)
(3) √(5gr) **(4)** gr

9. A spring compressed by x stores energy E. Compressed by 2x it stores

(1) 4E **(2)** 2E
(3) E/2 **(4)** 8E

10. A force of 100 N moves a body at a constant 5 m/s along the force. The power is

(1) 20 W **(2)** 105 W
(3) 500 W **(4)** 50 W

11. For a perfectly elastic collision, the coefficient of restitution is

(1) 0 **(2)** 0.5

(3) ∞ **(4)** 1

12. Torque equals the rate of change of

(1) linear momentum **(2)** angular momentum

(3) kinetic energy **(4)** angular velocity

13. The total kinetic energy of a disc of mass m rolling without slipping at speed v is

(1) ¾ mv^2 **(2)** ½ mv^2

(3) mv^2 **(4)** ¼ mv^2

14. The acceleration due to gravity at a depth R/2 below the Earth's surface is

(1) g/4 **(2)** 2g

(3) g/2 **(4)** zero

15. Kepler's second law follows from conservation of

(1) energy **(2)** linear momentum

(3) mass **(4)** angular momentum

16. Poisson's ratio has

(1) units of N/m^2 **(2)** no units and no dimensions

(3) units of m **(4)** units of N

17. The height of capillary rise of a liquid in a tube of radius r is proportional to

(1) 1/r **(2)** r

(3) r^2 **(4)** $1/r^2$

18. The flow of a liquid is streamline (laminar) when the Reynolds number is

(1) very large **(2)** infinite

(3) small (below about 1000) **(4)** negative

19. The coefficients of volume and linear expansion of an isotropic solid are related by

(1) $\gamma = \alpha$ **(2)** $\gamma = 2\alpha$

(3) $\gamma = \alpha/3$ **(4)** $\gamma = 3\alpha$

20. By Newton's law of cooling, the rate of cooling is proportional to the

(1) square of temperature **(2)** excess temperature over surroundings

(3) fourth power of temperature **(4)** mass only

21. Work done by a gas in an isochoric process is

(1) zero **(2)** $P\Delta V$

(3) $nR\Delta T$ **(4)** maximum

22. The efficiency of a Carnot engine working between 500 K and 300 K is

(1) 60% **(2)** 25%

(3) 40% **(4)** 50%

23. The average kinetic energy per molecule of an ideal gas at temperature T is

(1) ½kT **(2)** kT

(3) 3kT **(4)** (3/2)kT

24. A pendulum is taken to the Moon where g is one-sixth of Earth's. Its period becomes

(1) 6 times **(2)** √6 times

(3) 1/6 times **(4)** unchanged

25. In a stationary wave, the distance between adjacent nodes is

(1) $\lambda/2$ **(2)** λ

(3) $\lambda/4$ **(4)** 2λ

26. The fundamental frequency of a pipe of length L open at both ends is

(1) $v/4L$ **(2)** v/L

(3) $v/2L$ **(4)** $2v/L$

27. Quantisation of charge means q =

(1) $ne/2$ **(2)** e/n

(3) n^2e **(4)** ne

28. The electric field inside a charged hollow conducting sphere is

(1) maximum **(2)** zero

(3) equal to that at the surface **(4)** infinite

29. The work done in moving a charge along an equipotential surface is

(1) zero **(2)** positive

(3) negative **(4)** infinite

30. A dielectric slab is inserted between the plates of a capacitor kept connected to a battery. The charge on it

(1) decreases **(2)** stays the same

(3) increases **(4)** becomes zero

31. A wire of resistance R is stretched to double its length (volume constant). New resistance is

(1) R **(2)** 2R

(3) R/2 **(4)** 4R

32. Two cells of emf 1.5 V each are connected in series, aiding each other. Total emf is

(1) 1.5 V **(2)** 3 V

(3) 0.75 V **(4)** 0

33. Joule's law of heating gives heat produced as

(1) I^2Rt **(2)** IR^2t

(3) I^2R/t **(4)** IRt^2

34. The magnetic field at a distance r from a long straight wire carrying current I is

(1) $\mu_0 I/2r$ **(2)** $\mu_0 I/4\pi r$

(3) $\mu_0 I/2\pi r$ **(4)** $\mu_0 Ir/2\pi$

35. The cyclotron frequency of a charged particle is

(1) proportional to speed **(2)** proportional to radius

(3) inversely proportional to B **(4)** independent of speed

36. The angle of dip at the magnetic equator is

(1) 90° **(2)** 0°

(3) 45° **(4)** 60°

37. Magnetic flux through a single-turn coil changes from 0.5 Wb to 0.1 Wb in 0.2 s. The average induced emf is

(1) 2 V **(2)** 0.4 V

(3) 0.08 V **(4)** 4 V

38. The reactance of a capacitor, as frequency increases,

(1) increases **(2)** stays the same

(3) decreases **(4)** becomes infinite

39. Wattless current flows in a circuit when the power factor is

(1) 1 **(2)** 0.5

(3) 0.707 **(4)** 0

40. Which electromagnetic radiation has the shortest wavelength?

(1) X-rays **(2)** γ-rays

(3) UV rays **(4)** microwaves

41. Thin lenses of power +2 D and −0.5 D are in contact. The combined power is

(1) +1.5 D **(2)** +2.5 D

(3) −1.5 D **(4)** −1 D

42. Optical fibres work on the principle of

(1) refraction only **(2)** dispersion

(3) total internal reflection **(4)** diffraction

43. The resolving power of a microscope increases when the wavelength of light

(1) increases **(2)** is unchanged

(3) is doubled **(4)** decreases

44. The energy of a photon of wavelength 1240 nm is about (hc = 1240 eV nm)

(1) 1240 eV **(2)** 1 eV

(3) 10 eV **(4)** 0.1 eV

45. The universal logic gate among these is

(1) NAND **(2)** AND

(3) OR **(4)** NOT

Chemistry

46. The number of molecules in 18 g of water is

(1) 3.011×10^{23} **(2)** 1.8×10^{24}

(3) 6.022×10^{23} **(4)** 18

47. The molarity of a solution of 4 g NaOH in 500 mL is

(1) 0.2 M **(2)** 0.1 M

(3) 0.4 M **(4)** 2 M

48. The number of orbitals in the shell n = 2 is

(1) 2 **(2)** 8

(3) 6 **(4)** 4

49. According to Pauli's exclusion principle, an orbital can hold at most

(1) 1 electron **(2)** 2 electrons

(3) 6 electrons **(4)** 8 electrons

50. The electronic configuration of Cu (Z = 29) is

(1) [Ar] $3d^{10}\ 4s^{1}$ **(2)** [Ar] $3d^{9}\ 4s^{2}$

(3) [Ar] $3d^{8}\ 4s^{2}$ **(4)** [Ar] $3d^{10}$

51. Across a period from left to right, atomic radius generally

(1) increases **(2)** stays the same

(3) decreases **(4)** first decreases then increases sharply

52. The element with the lowest first ionisation enthalpy is

(1) Li **(2)** Na

(3) K **(4)** Cs

53. The H–C–H bond angle in CH_4 is

(1) 120° **(2)** 109.5°

(3) 107° **(4)** 104.5°

54. Which molecule is linear?

(1) CO_2 **(2)** H_2O

(3) SO_2 **(4)** NH_3

55. The unusually high boiling point of water is due to

(1) its small size **(2)** ionic bonding

(3) hydrogen bonding **(4)** van der Waals forces only

56. The N_2 molecule has

(1) 3 σ bonds **(2)** 2 σ and 1 π bond

(3) 1 σ and 1 π bond **(4)** 1 σ and 2 π bonds

57. At constant pressure, the volume of a fixed mass of gas is proportional to absolute temperature. This is

(1) Boyle's law **(2)** Charles' law

(3) Graham's law **(4)** Avogadro's law

58. For an ideal gas, the compressibility factor Z is

(1) 1 **(2)** 0

(3) greater than 1 always **(4)** less than 1 always

59. The first law of thermodynamics is

(1) $\Delta S = q/T$ **(2)** $\Delta G = \Delta H - T\Delta S$

(3) $\Delta U = q + w$ **(4)** $PV = nRT$

60. Hess's law follows from the fact that enthalpy is

(1) an intensive property **(2)** a path function

(3) always negative **(4)** a state function

61. A reaction is spontaneous at all temperatures when

(1) $\Delta H > 0, \Delta S < 0$ **(2)** $\Delta H < 0, \Delta S > 0$

(3) $\Delta H > 0, \Delta S > 0$ **(4)** $\Delta H < 0, \Delta S < 0$

62. For which reaction is Kp = Kc?

(1) $H_2 + I_2 \rightleftharpoons 2HI$ **(2)** $N_2 + 3H_2 \rightleftharpoons 2NH_3$

(3) $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ **(4)** $2SO_2 + O_2 \rightleftharpoons 2SO_3$

63. The pH of pure water at 25 °C is

(1) 0 **(2)** 14

(3) 7 **(4)** 1

64. The solubility of a sparingly soluble salt in the presence of a common ion

(1) increases **(2)** is unchanged

(3) becomes infinite **(4)** decreases

65. A reaction in which the same species is both oxidised and reduced is called

(1) combination **(2)** disproportionation

(3) displacement **(4)** decomposition

66. The structure of H_2O_2 is

(1) non-planar (open book) **(2)** linear

(3) planar **(4)** tetrahedral

67. Washing soda is

(1) $NaHCO_3$ **(2)** NaOH

(3) $Na_2CO_3 \cdot 10H_2O$ **(4)** Na_2SO_4

68. The anomalous behaviour of lithium is mainly due to its

(1) large size **(2)** low density

(3) high reactivity **(4)** small size and high polarising power

69. Thallium shows a stable +1 oxidation state due to

(1) high electronegativity **(2)** the inert pair effect

(3) large size only **(4)** d-electrons

70. Silicones are polymers containing repeating

(1) –Si–O–Si– units **(2)** –C–C– units

(3) –Si–Si– units **(4)** –Si–C–Si– units only

71. Hyperconjugation involves delocalisation of

(1) lone pairs **(2)** π electrons only

(3) σ electrons of C–H bonds **(4)** d electrons

72. The number of structural isomers of C_5H_{12} is

(1) 2 **(2)** 4

(3) 5 **(4)** 3

73. An alkyl halide treated with sodium in dry ether gives an alkane. This is the

(1) Kolbe reaction **(2)** Wurtz reaction

(3) Friedel–Crafts reaction **(4)** Sandmeyer reaction

74. Depletion of the ozone layer is mainly caused by

(1) CFCs **(2)** CO_2

(3) N_2 **(4)** water vapour

75. Which is a colligative property?

(1) refractive index **(2)** viscosity

(3) osmotic pressure **(4)** surface tension

76. A mixture that boils at a constant temperature with unchanged composition is a(n)

(1) ideal solution **(2)** colloid

(3) suspension **(4)** azeotrope

77. The van't Hoff factor for K_2SO_4 (complete dissociation) is

(1) 2 **(2)** 3

(3) 1 **(4)** 4

78. A galvanic cell converts

(1) chemical energy into electrical energy **(2)** electrical energy into chemical energy

(3) heat into work **(4)** light into electricity

79. On dilution, the conductivity (specific conductance) of an electrolyte solution

(1) increases **(2)** is unchanged

(3) decreases **(4)** becomes infinite

80. The half-life of a first-order reaction

(1) is proportional to initial concentration **(2)** is inversely proportional to concentration

(3) depends on the square of concentration **(4)** is independent of initial concentration

81. The Arrhenius equation is

(1) k = A e^(Ea/RT) **(2)** k = A e^(−Ea/RT)

(3) k = A Ea/RT **(4)** k = A RT/Ea

82. Zinc is not regarded as a transition element because

(1) its d-orbitals are completely filled in the atom and its common ion **(2)** it is a metal

(3) it has no electrons in s **(4)** it is very reactive

83. The lanthanoid that commonly shows a +4 oxidation state is

(1) La **(2)** Lu

(3) Ce **(4)** Gd

84. Ethylenediamine is a

(1) monodentate ligand **(2)** tridentate ligand

(3) ambident ligand **(4)** bidentate ligand

85. Chlorobenzene is less reactive towards nucleophilic substitution because the C–Cl bond has

(1) ionic character **(2)** partial double-bond character

(3) no polarity **(4)** a very long length

86. Williamson synthesis is used to prepare

(1) ethers **(2)** alcohols

(3) aldehydes **(4)** esters

87. Which compound gives a positive iodoform test?

(1) methanol **(2)** propanal

(3) acetone **(4)** benzaldehyde

88. Clemmensen reduction converts a carbonyl group into

(1) –CH(OH)– **(2)** –COOH

(3) –CH=CH– **(4)** $–CH_2–$

89. Aniline with $NaNO_2$/HCl at 0–5 °C gives

(1) nitrobenzene **(2)** benzenediazonium chloride

(3) phenol **(4)** benzamide

90. The α-helix is an example of protein

(1) secondary structure **(2)** primary structure

(3) tertiary structure **(4)** quaternary structure

Biology — Botany

91. A herbarium is a store of

(1) living plants
(2) dried, pressed and preserved plant specimens
(3) seeds only
(4) animal specimens

92. The taxonomic category just above genus is

(1) order **(2)** class
(3) family **(4)** species

93. Methanogens, found in the gut of ruminants, are

(1) eubacteria **(2)** cyanobacteria
(3) mycoplasma **(4)** archaebacteria

94. The cell walls of diatoms are made of

(1) cellulose **(2)** silica
(3) chitin **(4)** calcium carbonate

95. Bread mould Rhizopus belongs to

(1) Zygomycetes (Phycomycetes) **(2)** Ascomycetes
(3) Basidiomycetes **(4)** Deuteromycetes

96. 'Red tides' are caused by

(1) dinoflagellates such as Gonyaulax **(2)** diatoms
(3) euglenoids **(4)** slime moulds

97. The characteristic pigment of brown algae is

(1) phycoerythrin **(2)** chlorophyll b
(3) fucoxanthin **(4)** phycocyanin

98. In mosses, the dominant phase of the life cycle is the

(1) sporophyte **(2)** protonema only
(3) capsule **(4)** gametophyte

99. Heterospory is seen in

(1) Equisetum **(2)** Selaginella
(3) Pteris **(4)** Dryopteris

100. Coralloid roots of Cycas harbour

(1) cyanobacteria **(2)** mycorrhizal fungi
(3) Rhizobium **(4)** nematodes

101. Mango is an example of a

(1) berry **(2)** pome
(3) drupe **(4)** legume

102. Pneumatophores (respiratory roots) are found in

(1) Ficus **(2)** maize
(3) carrot **(4)** Rhizophora

103. The placentation in pea is

(1) axile **(2)** marginal
(3) parietal **(4)** free central

104. Onion (Allium) belongs to the family

(1) Liliaceae **(2)** Fabaceae
(3) Solanaceae **(4)** Poaceae

105. Collenchyma has cell walls thickened

(1) uniformly with lignin **(2)** with suberin
(3) at the corners with cellulose and pectin **(4)** with silica

106. Vessels are generally absent in the xylem of

(1) dicots **(2)** monocots
(3) angiosperms **(4)** gymnosperms

107. Casparian strips are present in the

(1) epidermis **(2)** endodermis
(3) pericycle **(4)** cortex

108. Heartwood is

(1) dark coloured and non-conducting **(2)** light coloured and conducting
(3) the outermost bark **(4)** living phloem

109. A cell organelle without a membrane is the

(1) mitochondrion **(2)** lysosome
(3) ribosome **(4)** vacuole

110. The nucleolus is the site of synthesis of

(1) DNA **(2)** mRNA only
(3) proteins **(4)** rRNA

111. Glycoproteins and glycolipids are formed in the

(1) ribosomes **(2)** Golgi apparatus
(3) nucleus **(4)** mitochondria

112. Centrioles show a

(1) 9 + 0 arrangement of triplets **(2)** 9 + 2 arrangement of doublets
(3) 9 + 3 arrangement **(4)** random arrangement

113. A chromosome with a terminal centromere is

(1) metacentric **(2)** submetacentric
(3) telocentric **(4)** acrocentric

114. The synaptonemal complex forms during

(1) leptotene **(2)** pachytene
(3) diplotene **(4)** zygotene

115. Chromosomes align at the cell's equator during

(1) prophase **(2)** metaphase
(3) anaphase **(4)** telophase

116. Plasmolysis occurs when a cell is placed in a

(1) hypertonic solution **(2)** hypotonic solution
(3) isotonic solution **(4)** pure water

117. Guttation takes place through

(1) stomata **(2)** lenticels
(3) hydathodes **(4)** cuticle

118. The element that is a constituent of chlorophyll is

(1) iron **(2)** calcium

(3) potassium **(4)** magnesium

119. The pink pigment in root nodules of legumes is

(1) haemoglobin **(2)** leghaemoglobin

(3) phycoerythrin **(4)** anthocyanin

120. Kranz anatomy is a feature of

(1) C_4 plants **(2)** C_3 plants

(3) CAM plants only **(4)** algae

121. Photorespiration involves

(1) only chloroplasts **(2)** chloroplast and mitochondria only

(3) chloroplast, peroxisome and mitochondria **(4)** only peroxisomes

122. Fixing one CO_2 in the Calvin cycle requires

(1) 2 ATP and 3 NADPH **(2)** 1 ATP and 1 NADPH

(3) 3 ATP and 3 NADPH **(4)** 3 ATP and 2 NADPH

123. The net gain of ATP (substrate level) in glycolysis per glucose is

(1) 4 **(2)** 2

(3) 36 **(4)** 8

124. Alcoholic fermentation by yeast produces

(1) ethanol and CO_2 **(2)** lactic acid

(3) acetic acid **(4)** pyruvate only

125. The final electron acceptor in aerobic respiration is

(1) NAD^+ **(2)** FAD

(3) oxygen **(4)** cytochrome c

126. Apical dominance is caused by

(1) cytokinin **(2)** ethylene

(3) ABA **(4)** auxin

127. The hormone that causes stomatal closure under water stress is

(1) gibberellin **(2)** abscisic acid

(3) auxin **(4)** cytokinin

128. Promotion of flowering by exposure to low temperature is

(1) vernalisation **(2)** photoperiodism

(3) phototropism **(4)** stratification

129. The innermost wall layer of the microsporangium is the

(1) endothecium **(2)** epidermis

(3) tapetum **(4)** middle layer

130. Pollination by water occurs in

(1) maize **(2)** Salvia

(3) sunflower **(4)** Vallisneria

131. A typical angiosperm embryo sac is

(1) 8-celled, 7-nucleate **(2)** 7-celled, 8-nucleate

(3) 8-celled, 8-nucleate **(4)** 7-celled, 7-nucleate

132. Crossing an F_1 individual with its recessive parent is a

(1) test cross **(2)** back cross with dominant parent
(3) reciprocal cross **(4)** dihybrid cross

133. The concept of linkage was established by experiments of

(1) Mendel **(2)** de Vries
(3) T. H. Morgan **(4)** Sutton

134. In bacteria, transcription of all RNA types is done by

(1) three RNA polymerases **(2)** DNA polymerase
(3) primase **(4)** a single RNA polymerase

135. A biodiversity hotspot in India is the

(1) Thar desert **(2)** Western Ghats
(3) Deccan plateau **(4)** Indo-Gangetic plain

Biology — Zoology

136. Pseudocoelomate animals belong to

(1) Aschelminthes **(2)** Annelida
(3) Chordata **(4)** Arthropoda

137. Flame cells are excretory structures in

(1) Annelida **(2)** Cnidaria
(3) Platyhelminthes **(4)** Mollusca

138. An open circulatory system is found in

(1) annelids **(2)** mammals
(3) cephalopods only **(4)** arthropods

139. A cartilaginous fish is

(1) Labeo **(2)** Scoliodon
(3) Catla **(4)** Hippocampus

140. Pneumatic bones are a feature of

(1) birds **(2)** reptiles
(3) amphibians **(4)** fishes

141. Areolar tissue is a type of

(1) epithelial tissue **(2)** muscular tissue
(3) loose connective tissue **(4)** nervous tissue

142. Cardiac muscle is

(1) striated and voluntary **(2)** unstriated and involuntary
(3) unstriated and voluntary **(4)** striated and involuntary

143. Excretion in cockroach is mainly by

(1) nephridia **(2)** Malpighian tubules
(3) green glands **(4)** flame cells

144. The excretory organs of earthworm are

(1) nephridia **(2)** Malpighian tubules
(3) kidneys **(4)** flame cells

145. Lecithin is a

(1) steroid **(2)** wax
(3) phospholipid **(4)** triglyceride

146. Malonate inhibits succinate dehydrogenase by

(1) denaturation **(2)** allosteric activation
(3) feedback stimulation **(4)** competitive inhibition

147. Pepsin acts best in a medium that is

(1) alkaline **(2)** acidic
(3) neutral **(4)** strongly basic

148. Villi in the small intestine

(1) increase the surface area for absorption **(2)** secrete bile
(3) store food **(4)** produce HCl

149. Kwashiorkor is caused by deficiency of

(1) vitamins **(2)** fats
(3) proteins **(4)** minerals only

150. The partial pressure of O_2 in alveolar air is about

(1) 40 mm Hg **(2)** 45 mm Hg
(3) 159 mm Hg **(4)** 104 mm Hg

151. Damage to alveolar walls reducing respiratory surface occurs in

(1) asthma **(2)** emphysema
(3) pneumonia **(4)** bronchitis

152. Erythroblastosis fetalis is due to

(1) Rh incompatibility **(2)** ABO incompatibility only
(3) vitamin K deficiency **(4)** iron deficiency

153. Compared to blood, lymph lacks

(1) WBCs **(2)** plasma proteins entirely
(3) RBCs **(4)** water

154. Blood passes through the heart twice in one cycle in mammals. This is

(1) single circulation **(2)** portal circulation
(3) open circulation **(4)** double circulation

155. Urea is synthesised in the

(1) kidney **(2)** liver
(3) pancreas **(4)** spleen

156. Renin is secreted by the

(1) juxtaglomerular cells **(2)** adrenal cortex
(3) pituitary **(4)** liver

157. The number of bones in the adult human skeleton is

(1) 208 **(2)** 212
(3) 206 **(4)** 201

158. Myasthenia gravis is an autoimmune disorder affecting the

(1) bones **(2)** joints
(3) kidneys **(4)** neuromuscular junction

159. Secretions of the pituitary are controlled by the

(1) thalamus **(2)** hypothalamus
(3) pons **(4)** cerebellum

160. Balance and posture are maintained by the

(1) cerebellum **(2)** medulla
(3) cerebrum **(4)** hypothalamus

161. Vision in dim light is due to

(1) cones **(2)** iodopsin
(3) rods (rhodopsin) **(4)** the lens

162. Cortisol is secreted by the

(1) adrenal medulla **(2)** thyroid
(3) pancreas **(4)** adrenal cortex

163. Cretinism results from

(1) hyperthyroidism in adults **(2)** hypothyroidism in children
(3) excess insulin **(4)** lack of GH in adults

164. The corpus luteum mainly secretes

(1) progesterone **(2)** FSH
(3) LH **(4)** testosterone

165. The acrosome of a sperm contains

(1) mitochondria **(2)** DNA
(3) enzymes that help penetrate the ovum **(4)** centrioles

166. hCG, a hormone of pregnancy, is secreted by the

(1) pituitary **(2)** ovary
(3) thyroid **(4)** placenta

167. Vasectomy is a

(1) barrier method **(2)** sterilisation method in males
(3) hormonal method **(4)** natural method

168. Klinefelter's syndrome has the karyotype

(1) 47, XXY **(2)** 45, X
(3) 47, XXX **(4)** 47, +21

169. Red–green colour blindness is

(1) autosomal dominant **(2)** Y-linked
(3) X-linked recessive **(4)** mitochondrial

170. In sickle-cell anaemia, the 6th amino acid of β-globin changes from

(1) valine to glutamic acid **(2)** lysine to valine
(3) glycine to alanine **(4)** glutamic acid to valine

171. Okazaki fragments are formed on the

(1) leading strand **(2)** lagging strand
(3) mRNA **(4)** tRNA

172. The adaptor molecule in protein synthesis is

(1) tRNA **(2)** mRNA
(3) rRNA **(4)** DNA

173. The Human Genome Project was completed in

(1) 1990 **(2)** 1998
(3) 2003 **(4)** 2010

174. Miller's experiment simulated

(1) cell division **(2)** photosynthesis
(3) natural selection **(4)** conditions of the primitive Earth

175. Industrial melanism is shown by

(1) Darwin's finches **(2)** Biston betularia (peppered moth)
(3) Drosophila **(4)** E. coli

176. Typhoid is confirmed by the

(1) Widal test **(2)** ELISA only
(3) Mantoux test **(4)** Schick test

177. Antibodies are produced by

(1) T-helper cells **(2)** macrophages
(3) B-lymphocytes (plasma cells) **(4)** neutrophils

178. Colostrum is rich in the antibody

(1) IgE **(2)** IgG
(3) IgM **(4)** IgA

179. PCR uses the thermostable enzyme

(1) DNA ligase **(2)** Taq polymerase
(3) restriction endonuclease **(4)** reverse transcriptase

180. The first clinical gene therapy was given for

(1) ADA deficiency **(2)** diabetes
(3) cancer **(4)** haemophilia

Answer Key

Physics

1. (3)	**2.** (1)	**3.** (4)	**4.** (2)	**5.** (1)	**6.** (3)
7. (4)	**8.** (2)	**9.** (1)	**10.** (3)	**11.** (4)	**12.** (2)
13. (1)	**14.** (3)	**15.** (4)	**16.** (2)	**17.** (1)	**18.** (3)
19. (4)	**20.** (2)	**21.** (1)	**22.** (3)	**23.** (4)	**24.** (2)
25. (1)	**26.** (3)	**27.** (4)	**28.** (2)	**29.** (1)	**30.** (3)
31. (4)	**32.** (2)	**33.** (1)	**34.** (3)	**35.** (4)	**36.** (2)
37. (1)	**38.** (3)	**39.** (4)	**40.** (2)	**41.** (1)	**42.** (3)
43. (4)	**44.** (2)	**45.** (1)			

Chemistry

46. (3)	**47.** (1)	**48.** (4)	**49.** (2)	**50.** (1)	**51.** (3)
52. (4)	**53.** (2)	**54.** (1)	**55.** (3)	**56.** (4)	**57.** (2)
58. (1)	**59.** (3)	**60.** (4)	**61.** (2)	**62.** (1)	**63.** (3)
64. (4)	**65.** (2)	**66.** (1)	**67.** (3)	**68.** (4)	**69.** (2)
70. (1)	**71.** (3)	**72.** (4)	**73.** (2)	**74.** (1)	**75.** (3)
76. (4)	**77.** (2)	**78.** (1)	**79.** (3)	**80.** (4)	**81.** (2)
82. (1)	**83.** (3)	**84.** (4)	**85.** (2)	**86.** (1)	**87.** (3)
88. (4)	**89.** (2)	**90.** (1)			

Biology — Botany

91. (2)	**92.** (3)	**93.** (4)	**94.** (2)	**95.** (1)	**96.** (1)
97. (3)	**98.** (4)	**99.** (2)	**100.** (1)	**101.** (3)	**102.** (4)
103. (2)	**104.** (1)	**105.** (3)	**106.** (4)	**107.** (2)	**108.** (1)
109. (3)	**110.** (4)	**111.** (2)	**112.** (1)	**113.** (3)	**114.** (4)
115. (2)	**116.** (1)	**117.** (3)	**118.** (4)	**119.** (2)	**120.** (1)
121. (3)	**122.** (4)	**123.** (2)	**124.** (1)	**125.** (3)	**126.** (4)
127. (2)	**128.** (1)	**129.** (3)	**130.** (4)	**131.** (2)	**132.** (1)
133. (3)	**134.** (4)	**135.** (2)			

Biology — Zoology

136. (1)	**137.** (3)	**138.** (4)	**139.** (2)	**140.** (1)	**141.** (3)
142. (4)	**143.** (2)	**144.** (1)	**145.** (3)	**146.** (4)	**147.** (2)
148. (1)	**149.** (3)	**150.** (4)	**151.** (2)	**152.** (1)	**153.** (3)
154. (4)	**155.** (2)	**156.** (1)	**157.** (3)	**158.** (4)	**159.** (2)
160. (1)	**161.** (3)	**162.** (4)	**163.** (2)	**164.** (1)	**165.** (3)
166. (4)	**167.** (2)	**168.** (1)	**169.** (3)	**170.** (4)	**171.** (2)
172. (1)	**173.** (3)	**174.** (4)	**175.** (2)	**176.** (1)	**177.** (3)
178. (4)	**179.** (2)	**180.** (1)			

Original model paper prepared for practice, following the latest announced exam pattern. Not an official NTA paper.